

Research article

Multi-Stage Optimization for Supporting Lines in Logistics

Companies

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Abstract: Route planning in logistics operations often relies on human experience, resulting in high operational costs and inconsistent routing quality. This study investigates the Vehicle Routing Problem with Time Windows (VRPTW) and proposes a Multi-stage Operationally Constrained VRP Framework based on predefined main routes provided by a logistics company. The framework sequentially performs store extraction, store assignment, and support route planning by integrating operational rules with stage-specific optimization methods, including the Cooperative Co-evolutionary Genetic Algorithm (CCGA), Hybrid Ant Colony Optimization (HACO), and Multiple Ant Colony System (MACS). Experimental results on 26 real-world delivery instances show that the proposed framework reduces the number of required routes by 7.00%, total travel distance by 3.07%, while improving route utilization by 6.70% and on-time delivery performance by 7.16% compared with manual planning. These results demonstrate the effectiveness of the proposed framework in enhancing routing quality and reducing operational costs.

Keywords: Vehicle Routing Problem, Genetic Algorithm, Ant Colony Optimization

1. Introduction

With the rapid expansion of e-commerce and the increasing heterogeneity of customer demands, efficient route planning has become a critical challenge for logistics companies aiming to enhance distribution efficiency while mitigating operational costs. The Vehicle Routing Problem (VRP) [1] is a well-known NP-hard optimization problem, making exact optimization approaches, such as Integer Linear Programming, computationally intractable for large-scale instances. Consequently, metaheuristic algorithms, including the Genetic Algorithm (GA) [2] and Ant Colony Optimization (ACO) [3] have been extensively employed to obtain high-quality solutions within acceptable computational time. More recently, hybrid metaheuristic approaches [4] have been developed to further enhance search efficiency and solution quality. Although numerous VRP variants, including the Capacitated Vehicle Routing Problem (CVRP) [5], the Vehicle Routing Problem with Time Windows (VRPTW) [6], and the Vehicle Routing Problem with Private Fleet and Common Carrier (VRPPC) [7], have been extensively investigated, most studies assume that delivery routes can be globally optimized. In contrast, the problem considered in this study is constrained by predetermined route structures, fixed service sequences, and regional operating constraints, which are rarely addressed in conventional VRP formulations.

To address these challenges, this study proposes a Multi-stage Operationally Constrained VRP framework that decomposes the planning process into three sequential stages consisting of store extraction, store assignment, and support route planning. By incorporating operational constraints into the optimization process, the proposed method reduces the search space while improving solution quality and computational efficiency. This paper makes the following contributions. First, a formal multi-stage VRP formulation under fixed main route structures is developed, decomposing the problem into subset selection, assignment, and routing subproblems. Second, a stage-specific optimization framework is designed to address different subproblem characteristics, improving both efficiency and solution quality. Third, a constraint-aware mathematical model is constructed to ensure practical applicability while maintaining consistency with real operational requirements, providing a foundation for the future development of delivery decision-making and intelligent logistics method.

This study is organized as follows. Section 2 reviews related work, while Section 3 details the proposed methodology and system design. The experimental setup and results are presented in Section 4. Finally, Section 5 concludes the paper and suggests directions for future research.

2. Related work

2.1 Vehicle Routing Problem (VRP)

The VRP is a NP-hard combinatorial optimization problem that aims to determine the optimal allocation of a homogeneous fleet of vehicles and the corresponding delivery routes according to customer demands while minimizing the overall cost under various operational constraints. As

practical logistics applications become increasingly complex, numerous VRP variants have been proposed to accommodate different operational requirements. Common variants include the Capacitated Vehicle Routing Problem (CVRP) [5], the Vehicle Routing Problem with Time Windows (VRPTW) [6], and the Vehicle Routing Problem with Private Fleet and Common Carrier (VRPPC) [7]. In the basic VRP model, the following constraints must be satisfied:

1. Each vehicle starts and ends its route at the depot.
2. Each customer is visited exactly once and served by exactly one vehicle.
3. The demand of every customer must be satisfied.

2.1.1 Capacitated Vehicle Routing Problem (CVRP)

The CVRP is a fundamental variant of the VRP. It considers a homogeneous fleet of capacitated vehicles that depart from a single depot to serve customers with known demands while satisfying vehicle capacity constraints. The mathematical formulation incorporates customer demand and vehicle capacity constraints while minimizing the transportation cost. Chen et al. [8] proposed an Iterated Variable Neighborhood Descent (IVND) algorithm based on Variable Neighborhood Descent (VND) [9], which enhances routing quality through multiple neighborhood search operators and a perturbation mechanism. To improve the scalability of CVRP, Alesiani et al. [10] proposed the Constrained Clustering Capacitated Vehicle Routing Solver (CC-CVRS).

2.1.2 Vehicle Routing Problem with Time Windows (VRPTW)

The VRPTW is one of the most widely studied variants of the VRP, in which time window constraints are incorporated in addition to vehicle capacity limitations. In the VRPTW, each customer must be served within a specified time interval while satisfying vehicle capacity constraints, with the objective of minimizing the transportation cost. The VRPTW has attracted extensive research attention, and a wide range of optimization algorithms have been developed to address this problem. Ombuki et al. [11] proposed a Multi-Objective Genetic Algorithm (MOGA) that minimizes both the number of vehicles and total travel distance using a Pareto ranking technique. Maroof et al. [12] developed a Hybrid Genetic Algorithm with Solomon Insertion Heuristic (HGA-SIH), which integrates random initialization with SIH and adopts integer encoding with decoding to ensure feasibility under capacity and time window constraints. SIH is also used as a local search operator to enhance convergence, and its effectiveness is validated on Solomon benchmark instances [13]. In addition, the Hybrid Heuristic Harmony Search Algorithm (HHHSA) [14] demonstrate that incorporating local search strategies can improve solution quality while maintaining computational efficiency.

2.1.3 Vehicle Routing Problem with Private Fleet and Common Carrier (VRPPC)

The VRPPC extends the classical VRP by introducing the option of outsourcing deliveries to external carriers in addition to using a private fleet. In this problem, when the capacity of the private

fleet is insufficient or service costs are excessively high, a subset of customer demands can be assigned to common carriers. The objective is to minimize the total transportation cost while balancing fixed vehicle costs, travel costs, and outsourcing costs. Côté et al. [15] proposed a Tabu Search (TS)-based heuristic that enables dynamic assignment decisions between private fleet and outsourced delivery, while incorporating neighborhood search operators and adaptive penalty mechanisms to handle capacity constraints. Similarly, Bolduc et al. [16] developed a perturbation-based metaheuristic that evaluates outsourcing decisions during both construction and improvement phases, effectively determining whether customers should be served by private fleet or external carriers to minimize overall cost.

2.2 Genetic Algorithm (GA)

GA, proposed by John Holland [2], is inspired by Darwin's theory of natural selection, simulating the principles of survival of the fittest and genetic inheritance, and has been widely applied to various optimization problems [17-19]. GA is a metaheuristic based on natural selection and genetic mechanisms for solving combinatorial optimization problems. In GA, each candidate solution is represented as an individual encoded by a chromosome composed of multiple genes. During the evolutionary process, individuals are selected according to their fitness values, while crossover and mutation operators are applied to generate new offspring. Crossover exchanges genetic information between individuals, and mutation introduces random modifications to maintain population diversity and prevent premature convergence. Beasley et al. [20] proposed a GA-based framework for the Multidimensional Knapsack Problem (MKP), incorporating dual-based profit estimation and a DROP/ADD repair operator to maintain feasibility while improving solution quality.

For large-scale optimization problems, the Cooperative Co-evolutionary Genetic Algorithm (CCGA) [21] extends the Cooperative Co-evolutionary Algorithm (CCEA) [22] by integrating GA with a divide-and-conquer framework. Adedigba et al. [23] applied CCGA to multi-robot task scheduling, where iterative vector clustering (iVec) is used to partition tasks based on spatial relationships with respect to a common depot. Each cluster is assigned to a subpopulation for independent evolution, while fitness evaluation is conducted cooperatively using representative individuals from other subpopulations to assess the solution quality.

2.3 Ant Colony Optimization (ACO)

ACO, proposed by Marco Dorigo [3], is a metaheuristic inspired by the foraging behavior of ants, in which pheromone trails are used to guide the search for shortest paths in combinatorial optimization problems. In the solution construction phase, each ant selects the next node using a roulette-wheel selection mechanism, where transition probabilities are determined by pheromone intensity and heuristic information, resulting in a sequential decision-making process suitable for routing problems. After solution construction, pheromone trails are updated based on solution quality to reinforce high quality solutions, while evaporation is applied to prevent premature convergence.

These mechanisms enable ACO to effectively address routing-related optimization problems and extend its applicability to complex constrained scenarios.

In extended ACO frameworks, Gambardella et al. [24] proposed MACS-VRPTW (Multiple Ant Colony System for Vehicle Routing Problems with Time Windows), which employs a multi-colony structure to simultaneously optimize the number of vehicles and total travel distance for multi-objective optimization. Bell et al. [25] introduced a multi-colony ACO framework for VRP, where each vehicle is associated with an independent pheromone structure to reduce route-level coupling, and combined with 2-opt local search, region-based pheromone updates, and candidate list strategies to improve solution quality and computational efficiency for large-scale instances.

Hybrid Ant Colony Optimization (HACO) aims to enhance the performance of Ant Colony Optimization (ACO) by addressing its limitations in balancing global exploration and local exploitation. The HACO framework integrates complementary heuristic strategies to improve search robustness and solution quality. Ding et al. [26] proposed an ACO variant combined with local search, incorporating a disaster operator to mitigate premature convergence and maintain diversity. For vehicle routing applications, Molina et al. [27] proposed a hybrid method combining Ant Colony System (ACS) and Memetic Algorithm for the Heterogeneous Vehicle Routing Problem with Time Windows (HVRPTW) problem, where ACS performs global construction and Variable Neighborhood Tabu Search (VNTS) is used for local search, in combination with a holding list mechanism to manage unserved customers, resulting in reduced routing cost and improved convergence behavior.

2.4 Comparison with existing VRP variants

2.4.1 Practical limitations of existing VRP models

VRP models and their extensions have been extensively studied for logistics optimization. However, most existing studies primarily focus on global optimization, while practical operational constraints are relatively less considered. Consequently, theoretically optimal solutions may be difficult to directly apply in real-world logistics environments. Existing VRP formulations and variants, such as CVRP and VRPTW, generally assume that delivery demands are known in advance and optimize customer sequences and vehicle assignments through global route reconstruction. In addition, VRPTW commonly permits vehicles to arrive before the beginning of a time window and wait until service is available. However, in the practical environment considered in this study, vehicles must arrive within the specified delivery period, and early arrival with waiting is not allowed, resulting in a more restricted feasible solution space.

However, the problem investigated in this study differs from these global optimization settings, as existing main-route structures must be preserved. In practice, main routes are established based on historical experience, and drivers are familiar with fixed store sequences. Therefore, unrestricted route reorganization, even with improved optimization results, may reduce practical acceptance. Furthermore, logistics companies typically prioritize utilizing remaining capacity of existing routes

before activating additional support routes, indicating a hierarchical decision process that is rarely represented in Existing VRP models. Although manual planning procedures can maintain operational feasibility through experience-based decisions, they are time-consuming and difficult to optimize globally when multiple routes require adjustment simultaneously. Therefore, existing VRP models still lack sufficient consideration of fixed route structures, operational preferences, and hierarchical dispatching decisions. This motivates the development of an optimization framework that balances practical feasibility and computational efficiency.

2.4.2 Multi-Stage Optimization Strategies for Complex VRP Problems

In addition to modeling limitations, practical logistics operations involve multiple constraints and hierarchical decision objectives, making it difficult for a single algorithm to effectively handle all decision stages. Therefore, multi-stage strategies have been increasingly adopted by decomposing complex problems into smaller subproblems and applying appropriate optimization methods according to problem characteristics. For VRP and its variants, the search space grows rapidly with the number of nodes, limiting the effectiveness of single-stage global optimization methods. Decomposition-based approaches can reduce problem complexity and improve solution efficiency by dividing the original problem into manageable components.

In VRP-related studies, Frías et al. [28] proposed a two-phase hybrid algorithm for the Energy Minimizing Vehicle Routing Problem by combining clustering methods, ACO, and local search to reduce problem complexity and improve solution quality. Similarly, Jiao et al. [29] developed a multi-stage vehicle routing algorithm based on task grouping for VRPEC, where problem decomposition and sequential optimization improved both computational efficiency and solution performance.

These studies demonstrate that multi-stage frameworks can effectively balance solution quality and computational efficiency for complex optimization problems. Based on this concept, this study integrates practical logistics planning procedures and decomposes the routing problem into multiple functional stages, where suitable optimization mechanisms are applied to each stage to obtain feasible and high-quality solutions within an acceptable computational time.

3. Research methodology

3.1 Approach framework

We propose a Multi-stage operationally constrained vehicle routing problem framework to address logistics route planning under practical operational constraints. As illustrated in **Figure 1**, the proposed framework consists of four modules: the Data Module, Route Module, API Module, and Evaluation Module. Each module performs a distinct function in the overall optimization process, enabling the framework to generate feasible and efficient delivery routes for real-world logistics operations.

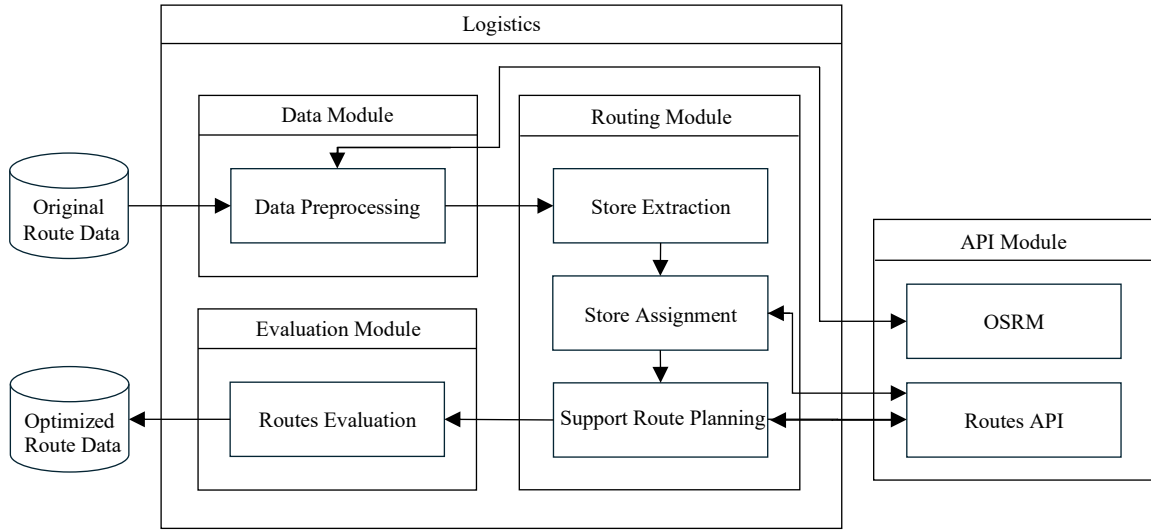


Figure 1. Research framework diagram

The Route Module serves as the core component of the proposed framework and is organized into three sequential stages. In the first stage, candidate stores are extracted from overloaded routes to form a candidate set for store assignment. In the second stage, candidate stores are assigned to main routes while satisfying operational constraints. In the third stage, support routes are constructed and optimized to improve overall routing quality. The remaining modules provide supporting functions for the routing process. The Data Module preprocesses the original route data and constructs a distance matrix using the Open Source Routing Machine (OSRM). The API Module utilizes the Google Maps Platform Routes API to estimate travel times, while the Evaluation Module evaluates the generated routing solutions and outputs the optimized routing results.

The proposed framework integrates multiple optimization algorithms to address the distinct objectives of different routing tasks. CCGA is adopted for candidate store extraction, HACO for store assignment, and MACS for support route planning and route optimization. By integrating these complementary algorithms, the proposed framework effectively addresses different routing tasks while maintaining operational feasibility and improving routing quality.

3.2 Problem Description and Operational Constraints

This study used the initial route data provided by the logistics company, which represented main delivery routes planned based on historical experience. These routes defined vehicle travel paths and store service sequences. The investigated problem focused on support route planning under existing main-route structures with volume and time window constraints. Due to daily demand fluctuations, route adjustments were required when main routes exceeded their capacity limits. To maintain practical feasibility, this study preserved the original store sequences and assigned overloaded stores to available or support vehicles, thereby improving delivery efficiency and reducing the number of required vehicles and transportation costs.

Operational constraints were derived from interviews with the logistics company to ensure feasible delivery planning. The constraints are defined as follows.

1. Load capacity: The maximum allowable load rate was 100%.
2. Store arrival time: The vehicle must arrive at each store within its specified time window.
3. Route duration: The working time of each route was limited to a maximum of 5 hours.
4. Regional and temporal: The store locations and scheduled arrival times are considered during route planning.
5. Main-route integrity: The main route can support candidate stores while preserving the original store sequence and minimizing store relocations to maintain operational familiarity.

The planning rules were established based on the analysis of historical route data and practical experience from the logistics company's route planning process. Accordingly, the following rules and procedures were summarized.

1. The volume of each main route was evaluated against the vehicle capacity limit. Routes exceeding the limit underwent store extraction, and the extracted stores were included in the candidate store set.
2. The candidate stores were geographically grouped to support subsequent regional support route planning.
3. Candidate stores were assigned to main routes with available capacity while satisfying capacity and time window constraints. The impact on route duration was then evaluated.
4. Unassigned candidate stores were planned as support routes based on their distance from the logistics center. Stores were classified into near, middle, and distant regions, where distant-region routes maintained directional consistency to avoid unnecessary detours while satisfying constraints.
5. The road network distances and travel times were estimated using OSRM during route planning. The completed routes were further evaluated using the Google Maps Routes API to obtain estimated travel distances and static durations. The updated route information was then integrated for subsequent evaluation.

3.2.1 Mathematical model

1. Variable definitions:

- Q_k : Maximum volume capacity of vehicle k .
- Q_{sup} : Maximum volume capacity of a support vehicle.
- L_k : Total volume assigned to vehicle k .
- v_i : Volume demand of store i .
- T_{max} : Maximum route duration of a vehicle.
- T_k : Total route duration of vehicle k .
- $d_{0,j}$: Distance between the logistics center and store j .
- $d_{i,j}$: Distance between store i and store j .

e_i : Earliest service time of store i .
 l_i : Latest service time of store i .
 $t_{i,j}$: Travel time between store i and store j .
 s_i : Service time of store i .
 C : Fixed transportation cost per vehicle.

2. Set definitions:

M : Set of main routes.
 H : Set of support routes.
 N : Set of all stores.
 K : Set of all vehicles.
 R : Set of candidate stores.
 R_{near} : Set of near-area candidate stores.
 R_{mid} : Set of mid-area candidate stores.
 R_{far} : Set of far-area candidate stores.

3. Genetic algorithm notation definitions:

P_k : k -th subpopulation.
 P_k^g : k -th subpopulation in generation g .
 N_{sub} : Size of each subpopulation.
 E : Number of elite individuals.
 G : Maximum number of generations.
 $F_{k,n}^g$: Fitness value of the n -th individual in the k -th subpopulation at generation g .
 $p_{k,n}^g$: Selection probability of the n -th individual in the k -th subpopulation at generation g .
 $Z_{k,n}^g$: n -th individual in the k -th subpopulation at generation g .
 p_c : Crossover probability.
 p_m : Mutation probability.
 F : Fitness function.
 w_1 : Vehicle penalty coefficient.
 w_2 : Store count penalty coefficient.
 w_i : Weight coefficient of store i .
 θ_i : Demand level coefficient of store i .

4. Ant colony optimization notation definitions:

α : Relative importance of pheromone information.
 β : Relative importance of heuristic information.
 ρ : Pheromone evaporation rate.
 m : Number of ants.
 τ_0 : Initial pheromone concentration.
 $\tau_{i,j}$: Pheromone concentration between store i and store j .
 $\Delta\tau_{i,j}^{iter}$: Quantity of pheromone laid on edge (i, j) by ant k .

$\eta_{i,j,u,k}$: Heuristic information for inserting store u between stores i and j in route k .
 $\eta_{i,j}$: Distance heuristic information.
 $p_{i,j,u,k}^a$: Transition probability for ant a to insert store u between stores i and j in route k .
 $p_{i,j}^a$: Transition probability for ant a to move from store i to store j .
 N_i^a : Feasible neighborhood set available to ant a at store i .
 N_k : k -th neighborhood structure.
 $\Phi_{i,j}$: Travel cost from store i to store j
 q : Random variable.
 q_0 : Greedy selection threshold.

4. Decision variables:

$z_{k,n,i}^g \in \{0,1\}$: Extraction decision of store i by the n -th individual in the k -th subpopulation at generation g .
 $x_{i,j}^k \in \{0,1\}$: Binary variable indicating whether vehicle k travels from store i to store j .

5. Constraints:

Equation (3.1) represents the constraint that each store is visited exactly once by a vehicle.

$$\sum_{k \in K} \sum_{i \in N} x_{i,j}^k = 1, \quad \forall j \in N \quad (3.1)$$

Equations (3.2) and (3.3) represent the departure and return constraints, respectively, ensuring that each vehicle starts and ends its route at the logistics center.

$$\sum_{j \in N} x_{0,j}^k = 1, \quad \forall k \in K \quad (3.2)$$

$$\sum_{i \in N} x_{i,0}^k = 1, \quad \forall k \in K \quad (3.3)$$

Equation (3.4) represents the vehicle capacity constraint, ensuring that the total volume assigned to each vehicle does not exceed its volume capacity.

$$\sum_{i \in r_k} v_i \leq Q_k, \quad \forall k \in K \quad (3.4)$$

Equation (3.5) represents the arrival time constraint, ensuring that a vehicle can arrive at the next store only after completing service at the previous store and traveling between the two stores.

$$t_j \geq t_i + s_i + t_{i,j}, \quad \forall i, j \in N \quad (3.5)$$

Equation (3.6) represents the time window constraint, ensuring that each store is served within its specified service time window. In this study, the time window is defined as one hour before to thirty minutes after the scheduled arrival time.

$$e_j \leq t_j \leq l_j, \quad \forall j \in N \quad (3.6)$$

Equation (3.7) represents the route duration constraint, ensuring that the total route duration of each vehicle does not exceed the maximum route duration of five hours.

$$T_k \leq T_{max}, \quad \forall k \in K \quad (3.7)$$

Equations (3.8), (3.9), and (3.10) represent the regional classification constraints, where candidate stores are categorized based on their distances from the logistics center into near, middle, and far regions.

$$R_{near} \Rightarrow d_{0,j} \leq 3, \quad \forall j \in N \quad (3.8)$$

$$R_{mid} \Rightarrow 3 < d_{0,j} \leq 5, \quad \forall j \in N \quad (3.9)$$

$$R_{far} \Rightarrow d_{0,j} > 5, \quad \forall j \in N \quad (3.10)$$

5. Objective function:

This study formulates a lexicographical optimization model for logistics route planning to improve operational efficiency. The model sequentially minimizes the number of support vehicles and total travel distance. The mathematical formulation is presented in Equation (3.11), with the two objective functions defined in Equations (3.12) and (3.13), respectively.

$$\text{Lex-Min} (f_{vehicle}(S), f_{travel}(S)) \quad (3.11)$$

$$f_{vehicle}(S) = \sum_{h \in H} \sum_{\substack{j \in N \\ (j \neq 0)}} x_{0,j}^h \quad (3.12)$$

$$f_{travel}(S) = \sum_{k \in K} \sum_{i \in N} \sum_{\substack{j \in N \\ (j \neq i)}} d_{i,j} \cdot x_{i,j}^k \quad (3.13)$$

3.3 Cooperative co-evolutionary genetic algorithm

CCGA was adopted for store extraction due to the decomposable characteristics of the problem. The store extraction decisions allowed the problem to be decomposed into multiple subproblems. Each subpopulation evolved partial solutions independently, and the cooperative evaluation mechanism combines them to assess the solution quality.

The principles and implementation process of CCGA were described as follows.

1. Encoding:

The store extraction problem was represented by a binary vector, where 1 indicated store extraction into the candidate set and 0 represented retention in the original route. Equation (3.14) presents the binary solution representation.

$$Z_{k,n}^g = [z_{k,n,1}^g, z_{k,n,2}^g, \dots, z_{k,n,|M_k|}^g] \quad (3.14)$$

2. Selection:

Roulette wheel selection is adopted for parent selection based on fitness values. Individuals with higher fitness values have higher selection probabilities, while elitism preserves the best individuals to prevent solution degradation. Equation (3.15) presents the selection probability calculation.

$$p_{k,n}^g = \frac{F_{k,n}^g}{\sum_{r=1}^{N_{sub}} F_{k,r}^g} \quad (3.15)$$

3. Crossover:

After selection, uniform crossover is performed within each subpopulation. Parents are either directly copied or recombined through gene exchange according to the crossover probability to generate offspring.

4. Mutation:

After crossover, Bit Flip Mutation is applied to offspring individuals. According to the mutation probability, selected gene values are flipped between 0 and 1 to maintain population diversity and prevent premature convergence.

5. Fitness:

The fitness value is evaluated using a cooperative evaluation mechanism. Since each subpopulation represents only a partial solution, context solutions from other subpopulations are incorporated to construct a complete solution for evaluation. A greedy algorithm and a nearest neighbor algorithm are adopted for candidate store assignment and support route planning, respectively. The fitness function is defined as a weighted linear combination of the number of support routes and total travel distance, as presented in Equation (3.16).

$$F(S) = w_1 * f_{vehicle}(S) + w_2 \cdot |R| + f_{travel}(S) \quad (3.16)$$

6. Repair operator:

Since crossover and mutation may violate solution feasibility, a heuristic repair mechanism is introduced to maintain feasible solutions. The mechanism uses a cost-based decision process to determine stores removed from overloaded routes. Extraction weights are calculated for each store, as shown in Equation (3.17).

$$w_i = \frac{C}{n} + \theta_i \cdot d_{0,i} \quad (3.17)$$

3.4 Hybrid ant colony optimization

HACO is adopted as the search mechanism for candidate store assignment to determine the allocation of candidate stores to main routes. HACO utilizes pheromone and heuristic information for solution construction and candidate solution generation. Furthermore, VND is incorporated as a local improvement procedure to improve solutions through multi-neighborhood search.

The principles and implementation process of HACO are described as follows.

1. Candidate store insertion mechanism:

The store selection and route construction process considers pheromone information, heuristic information, and the current route structure to determine the route and insertion position of each unassigned candidate store. Equation (3.18) defines the insertion probability, while Equation (3.19) represents the heuristic information, including insertion distance and

load costs. Equation (3.20) defines the insertion distance cost, Equation (3.21) evaluates the relationship between remaining vehicle capacity and store demand, and Equation (3.22) defines the weighting constraints of the heuristic components.

$$p_{i,j,u,k}^a = \frac{\tau_{i,u}^\alpha \cdot \eta_{i,j,u,k}^\beta}{\sum_{z \in N_i^a} \tau_{i,z}^\alpha \cdot \eta_{i,j,z,k}^\beta} \quad (3.18)$$

$$\eta_{i,j,u,k} = \frac{1}{\alpha_1 \cdot C_{i,j,u}^0 + \alpha_2 + C_{i,j,u}^1} \quad (3.19)$$

$$C_{i,j,u}^0 = d_{i,u} + d_{u,j} - d_{i,j} \quad (3.20)$$

$$C_{i,j,u}^1 = Q_k - L_k - v_u \quad (3.21)$$

$$\alpha_1 + \alpha_2 = 1 \quad (3.22)$$

2. Pheromone initialization and update:

The objective function guided ants toward high-quality solutions based on the total travel distance of support routes. Equation (3.23) defines the initial pheromone value based on the greedy solution. Equation (3.24) represents the local pheromone update to maintain search diversity, while Equation (3.25) defines the global pheromone update to reinforce best solution paths and avoid premature convergence. Equation (3.26) defines the pheromone increment based on the current best solution.

$$\tau_0 = \frac{1}{|R| \cdot F(S_{gr})} \quad (3.23)$$

$$\tau_{i,j} = (1 - \rho) \cdot \tau_{i,j} + \rho \cdot \tau_0 \quad (3.24)$$

$$\tau_{i,j} = (1 - \rho) \cdot \tau_{i,j} + \rho \cdot \Delta\tau_{i,j} \quad (3.25)$$

$$\Delta\tau_{i,j} = \begin{cases} \frac{1}{F(S)} & \text{if the edge } (i,j) \in S, \\ 0 & \text{otherwise,} \end{cases} \quad (3.26)$$

3. Variable neighborhood descent:

VND was applied to improve the solution quality of candidate store assignment based on the initial solution. Four neighborhood structures, including 2-opt, Swap, Relocate, and Cross exchange, were sequentially explored to optimize route configurations. To preserve the original main-route structure, VND was only applied to assigned candidate stores. Equation (3.27) defines the new solution set generated by applying the k-th neighborhood structure to the current solution.

$$S^* = \arg \min_{S' \in N_k(S)} F(S') \quad (3.27)$$

4. Objective function:

The objective function of HACO was used to evaluate the overall cost of candidate store assignments, as formulated in Equation (3.28). In this stage, greedy based assignment evaluation was not adopted, and candidate store assignments were directly optimized.

$$F(S) = w_1 * f_{vehicle}(S) + f_{travel}(S) \quad (3.28)$$

3.5 Multiple ant colony system

MACS was adopted for support route planning and integrated with VND for local solution improvement. In MACS, different ant colonies optimize distinct objectives, where ACS-VEI minimizes the number of vehicles and ACS-DIST minimizes the total travel distance. MACS performs global exploration, while VND further improves solution quality through multi-neighborhood search.

The principles and implementation process of MACS are described as follows.

1. ACS-VEI:

The first colony in the MACS framework focused on minimizing the number of vehicles. It improved customer coverage efficiency during route construction by generating routes through an incremental insertion mechanism and prioritizing less frequently selected customer nodes using a memory mechanism.

2. ACS-DIST:

The second colony in the MACS framework focused on minimizing the total travel distance under a fixed number of vehicles. The solution construction process generated delivery routes, and VND was applied to improve solution quality. At the end of each iteration, candidate solutions were compared with the global best solution to update the best solution and pheromone distribution.

3. Store transition probability:

The store selection and route construction process calculated candidate store selection probabilities based on pheromone concentration and distance heuristic information. Equation (3.29) defines the selection probability, while Equations (3.30) and (3.31) define the distance heuristic information and travel cost, respectively. A memory term was incorporated to increase the selection probability of less frequently selected stores and improve search diversity.

$$p_{i,j}^a = \frac{\tau_{i,j} \cdot \eta_{i,j}^\beta}{\sum_{z \in N_i^a} \tau_{i,z} \cdot \eta_{i,z}^\beta} \quad (3.29)$$

$$\eta_{i,j} = \frac{1}{\Phi_{i,j}} \quad (3.30)$$

$$\Phi_{i,j} = \max(1.0, d_{i,j} - IN_j) \quad (3.31)$$

4. Pheromone initialization and update

The pheromone initialization and update mechanisms in the support route planning stage followed the same design as HACO. Equation (3.23) defines the initial pheromone value. Local and global pheromone updates were performed using Equations (3.24) and (3.25),

respectively, to maintain search diversity and enhance high-quality paths. Equation (3.26) defines the pheromone increment based on the objective value.

5. Variable neighborhood descent

VND was integrated into ACS-DIST to improve support route planning. Since ACS-DIST may converge to local optima due to pheromone accumulation, VND was applied to enhance solution quality. The VND procedure followed the neighborhood search mechanism described in HACO, and the k -th neighborhood solution set was defined in Equation (3.27).

4. Experimental results and analysis

4.1 Experimental environment

The specifications of the experimental equipment used in this study are shown in **Table 1**.

Table 1. Experimental equipment specifications.

Component	Specification
OS	Windows 11
CPU	Intel® Core™ i7-13700 5.20 GHz
GPU	Intel® UHD graphics 770
RAM	24GB
HDD	1TB SSD

4.2 Evaluation criteria

This study adopts the manually planned results provided by the logistics company as the benchmark for evaluating the effectiveness of the proposed method in practical logistics route planning. Since the generated routes are reviewed by the logistics company to verify their operational feasibility, such reviews focus on practical feasibility but do not provide a quantitative evaluation of routing performance. Therefore, the total number of routes, total travel distance, total travel time, average route loading rate, on-time rate, and planning time are adopted as quantitative performance metrics.

The number of routes represents the required vehicles for daily delivery operations, where fewer routes indicate lower costs. Total travel distance and travel time were used to evaluate transportation efficiency, with shorter values indicating improved routing performance under the same vehicle conditions. Average route loading rate measured vehicle capacity utilization, while on-time rate evaluated delivery service quality based on compliance with designated time windows. Planning time was used to assess computational efficiency, where shorter planning time indicated higher planning efficiency.

4.3 Dataset

In this study, route and store data obtained from a logistics company were employed. The dataset comprised original routes representing the company’s established main routes, each consisting of 3 to 15 stores. Since the collected data were manually planned routes generated by the logistics company, data preprocessing was first conducted before applying the subsequent optimization procedures. The experimental dataset contained manually planned route data from 26 operational days. The number of main routes for each day ranged from 81 to 88, with most cases containing approximately 81 main routes. Each route included approximately 3 to 15 stores, and the total number of stores served per day ranged from 717 to 769. The composition of daily delivery stores, store locations, and demand volumes varied according to order requirements. **Figure 2** illustrates the distribution of route loading rates across different operational days.

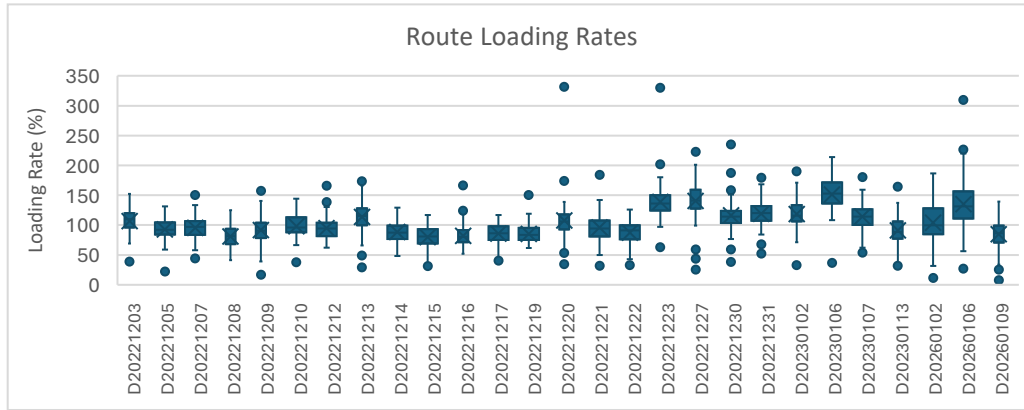


Figure 2. Distribution of route loading rates for original route instances.

4.4 Parameter settings and calibration

Table 2. Parameter settings of the proposed framework.

Method	Parameters
CCGA	Generation=200, Subpopulation size=20, Crossover rate=0.8 Mutation rate=0.1, Early stopping patience=40
HACO	Iteration=200, Number of Ants=50, Evaporation rate=0.7 Heuristic factor=1, Early stopping patience=40
MACS	Time limit=100, Number of Ants=50, Evaporation rate=0.5 Heuristic factor=7

4.5 End-to-end route planning pipeline implementation

4.5.1 Data preprocessing

The data preprocessing stage standardized and integrated the data based on route and store IDs to establish delivery routes and store lists. Missing values, abnormal records, and duplicate data were corrected to ensure data consistency. Store information, including geographic coordinates, time

windows, volume, and service time, was organized for subsequent route planning and algorithm execution.

4.5.2 Store Extraction

After data preprocessing, the store extraction stage was performed to resolve overloaded routes caused by demand fluctuations. An initial population was generated, where each individual represented an extraction decision from overloaded routes. The CCGA process then applied selection, crossover, and mutation operations, with fitness evaluated based on total route distance and vehicle cost. The best individual and extracted store list were obtained for subsequent store assignment. The pseudocode is presented in **Figure 3**.

01	Begin
02	Initialize context solution $b = \emptyset$
03	For each route r_k in M Do
04	Initialize subpopulation P_k^0
05	Evaluate the fitness of each individual in P_k^0 combined with b
06	Update b based on the best individual in P_k^0
07	End For
08	For each generation $g = 0$ to G Do
09	For each route r_k in M Do
10	Preserve the top E individuals from P_k^g and retain to P_k^{g+1}
11	While $ P_k^{g+1} < N_{sub}$ Do
12	$n = P_k^{g+1} + 1$
13	Select parents $Z_{k,a}^g, Z_{k,b}^g$ from P_k^g based on fitness
14	Perform Crossover to produce offspring $Z_{k,n}^{g+1}, Z_{k,n+1}^{g+1}$
15	Mutate offspring $Z_{k,n}^{g+1}, Z_{k,n+1}^{g+1}$
16	Add offspring $Z_{k,n}^{g+1}, Z_{k,n+1}^{g+1}$ to the new population P_k^{g+1}
17	End While
18	Evaluate the fitness of each individual in P_k^{g+1} combined with b
19	Update b based on the best individual in P_k^{g+1}
20	End For
21	End For
22	End

Figure 3. Pseudocode of CCGA.

4.5.3 Store Assignment

This stage assigned candidate stores to main routes with available capacity under volume and time window constraints to maximize route utilization and reduce support route requirements. HACO

was adopted for candidate store assignment, where a greedy strategy generated the initial solution for pheromone initialization, and multiple ant solutions were constructed in each iteration. VND was applied to the best solution after each iteration to further improve solution quality. The pseudocode is presented in **Figure 4**.

01	Begin
02	Generate and evaluate solution S_{gr}
03	Initialize pheromone τ_0
04	While stopping criterion is not met Do
05	Initialize solution $S_{iter} = \emptyset$
06	For each ant m Do
07	S_m = Solution Construction
08	If $\text{cost}(S_m) < \text{cost}(S_{iter})$ Then
09	$S_{iter} = S_m$
10	End If
11	End For
12	Apply VND to improve the solution S_{iter}
13	Update global pheromone $\tau_{i,j}$ based on S_{best}
14	End While
15	End

Figure 4. Pseudocode of HACO.

4.5.4 Support Route Planning

After store assignment, the support route planning stage was conducted to construct additional routes for stores that could not be reassigned to main routes due to vehicle capacity and time constraints. MACS was adopted for route construction and optimization, where ACS-VEI minimized the number of support vehicles and ACS-DIST further reduced the total travel distance under a fixed number of vehicles. The complete pseudocode of MACS is presented in **Figure 5**.

01	Begin
02	Generate and evaluate solution S_{nn} using nearest neighbor heuristic
03	Initialize pheromone τ based on solution S_{nn}
04	Initialize $S_{best} = S_{nn}$
05	While stopping criterion is not met Do
06	$k = S_{best} $
07	Activate ACS-VEI($k - 1$)
08	Activate ACS-DIST(k)
09	While ACS-VEI and ACS-DIST are active Do
10	Wait an improved solution S_{opt} from ACS-VEI or ACS-DIST

11	$S_{best} = S_{opt}$
12	If $ S_{best} < k$ Then
13	Kill ACS-VEI and ACS-DIST
14	End If
15	End While
16	End While
17	End

Figure 5. Pseudocode of MACS.

4.5.5 Performance Evaluation

This section integrated the routes generated by the proposed method and evaluated their performance. The main and support routes were combined to form the final plan, and Google Routes API was applied to obtain travel distances and estimated travel times. The results were evaluated using quantitative metrics, including vehicle number, total travel distance, average route loading rate, and on-time rate, for comparing different planning strategies.

4.6 Results and analysis

4.6.1 Notation Definitions

Ins.: Logistics instance index.
NV: Number of vehicles.
TD: Total distance.
U: Average route loading rate.
O: On-time rate.

4.6.2 Comparison of Experimental Results

The proposed method was evaluated using manually planned routes provided by the logistics company as the baseline over 26 daily instances. Performance was evaluated in terms of the number of vehicles, total travel distance, average route loading rate, on-time rate, and planning time. **Table 3** and **Figure 6** present the experimental results, including detailed comparisons of all evaluation metrics, the average improvements achieved by the proposed method, and the planning time of manual planning, the baseline method, and the proposed method.

Table 3. Comparison between the proposed method and logistics company planning.

<i>Ins.</i>	Proposed Method				Manually Planned			
	<i>NV</i>	<i>TD</i>	<i>U</i> (%)	<i>O</i> (%)	<i>NV</i>	<i>TD</i>	<i>U</i> (%)	<i>O</i> (%)
D20221203	93.4	4,508.47	92.3	99.87	102	4,623.34	84.85	94.7
D20221205	83.4	3,951.71	89.99	98.62	90	4,069.95	83.62	94.04

D20221207	85.7	4,093.01	90.65	99.56	89	4,134.37	86.95	90.38
D20221208	82	3,798.70	79.72	98.61	83	3,826.14	78.76	89.89
D20221209	84.2	3,952.41	88.34	99.69	89	4,042.17	83.4	89.46
D20221210	85.2	4,100.60	93	99.79	94	4,264.06	85.26	91.47
D20221212	83.4	3,986.38	91.43	99.94	91	4,169.74	85.74	92.93
D20221213	99.7	4,719.31	93.13	99.76	108	4,835.70	85.15	96.26
D20221214	82.5	3,906.81	86.86	99.29	85	3,915.92	84.22	92.4
D20221215	81	3,796.71	79.66	98.06	81	3,807.76	79.6	90.04
D20221216	81	3,802.63	82.13	97.49	83	3,859.83	80.19	87.36
D20221217	81	3,791.13	85.66	98.67	83	3,855.24	83.59	90.18
D20221219	81	3,805.58	84.93	97.34	82	3,812.78	83.93	88
D20221220	92.7	4,416.36	93.95	99.67	100	4,542.96	87.09	93.78
D20221221	85.5	4,099.36	90.82	98.71	90	4,172.23	86.29	91.5
D20221222	81.7	3,837.39	86.8	98.61	84	3,882.31	84.31	88.86
D20221223	121.5	5,591.63	92.66	99.77	129	5,567.91	87.21	95.2
D20221227	123	5,662.11	92.53	99.78	134	5,726.80	85.12	95.75
D20221230	101.2	4,760.85	92.53	99.83	105	4,701.75	89.56	94.95
D20221231	103.5	4,907.46	92.27	99.95	111	4,946.66	86.76	94.13
D20230102	103.7	4,973.15	92.51	99.74	110	4,883.74	87.05	96.52
D20230107	98.7	4,716.04	93.13	99.83	120	5,373.11	76.9	91.75
D20230113	83.2	3,919.75	88.24	98.44	91	4,168	80.88	91.96
D20260102	99.5	4,606.42	91.9	98.17	136	5,553.82	67.91	94.15
D20260106	128.7	6,011.78	92.3	99.74	159	6,891.41	74.62	93.11
D20260109	88.8	3,996.55	83.74	95.82	94	4,191.59	79.06	94.27



Figure 6. Planning time of the proposed method for each instance.

The experimental results demonstrate that the proposed method effectively improves delivery efficiency while reducing logistics costs. Compared with manual planning, the proposed method

reduced the number of vehicles by an average of 7.00%, total travel distance by 3.07%, and increased the average route loading rate and on-time rate by 6.70% and 7.16%, respectively. In addition, the planning time was reduced from approximately 1–3 hours to 2–30 minutes, significantly improving planning efficiency. These results demonstrate the effectiveness of the proposed method in enhancing operational efficiency and service quality in practical logistics operations. The details of the comparison methods are presented in **Table 4**.

4.6.3 Ablation Study and Performance Comparison

An ablation study was conducted to evaluate the contribution of each mechanism in the proposed multi-stage delivery planning framework. Three variants, including Random Extraction, HACO without VND, and MACS without VND, were compared under the same instances and parameter settings to ensure fairness. In addition, HGA-SIH was adopted as a baseline to compare the performance differences between multi-stage and single-stage optimization approaches. These comparisons further validated the effectiveness of each module and the overall framework.

Table 4. Description of Comparison Methods

Method	Description
HGA-SIH[12]	Single-stage optimization baseline
Random Extraction	CCGA-based store extraction replaced by random selection
HACO Without VND	HACO without VND local search in store assignment
MACS Without VND	MACS without VND local search in support route planning
Proposed Method	Multi-stage operationally constrained VRP framework

Figure 7 show that the proposed multi-stage optimization method achieved better performance than Single-Stage GA in most test cases across all evaluation metrics. This indicates that the problem decomposition and stage-wise optimization strategy can effectively reduce delivery costs and improve overall planning quality. Since the proposed method and the three ablation variants are based on the same multi-stage framework, their performance trends are relatively similar. The comparison results indicate that removing the CCGA-based candidate store extraction mechanism substantially affected the number of vehicles and total travel distance, highlighting the importance of effective store extraction from overloaded routes for subsequent route optimization. Moreover, excluding the VND local search strategy from HACO and MACS resulted in increased vehicle requirements and travel distances, demonstrating the effectiveness of VND in refining route structures and improving overall delivery efficiency.

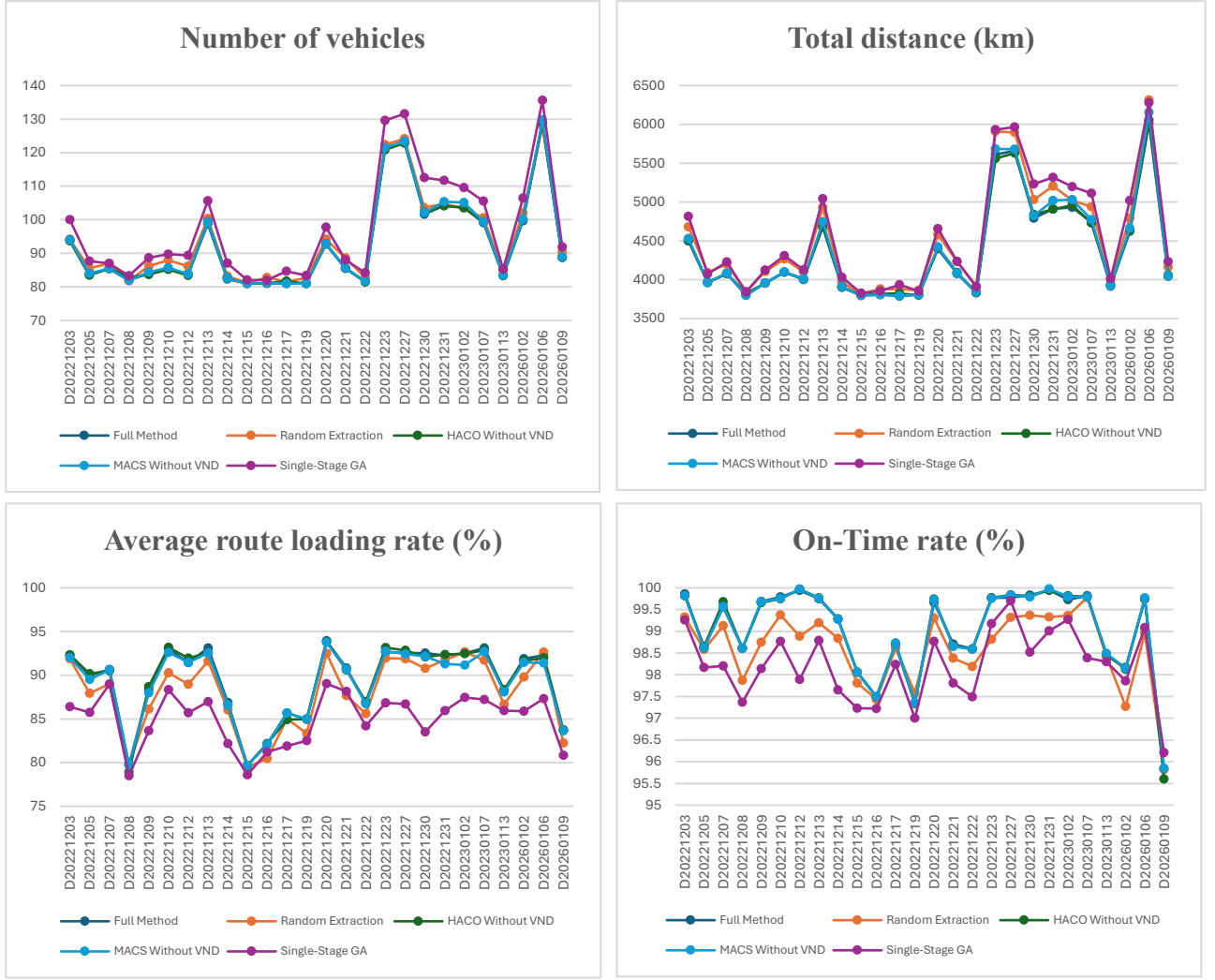


Figure 7. Comparison of routing performance for different instances

5. Conclusions

This study investigates the route optimization problem in practical logistics operations under a fixed main route structure. Considering the constraints of real-world logistics environments, including fixed store sequences in main routes, demand fluctuations causing route overloads, and the need for additional support routes, this study formulates the problem as a Multi-Stage Operationally Constrained Vehicle Routing Problem.

To address the complexity of the problem, the proposed framework decomposes the routing process into three sequential stages, including overloaded store extraction, candidate store assignment, and support route planning. Different optimization algorithms are designed according to the characteristics of each stage, where CCGA is applied for candidate store extraction, HACO is utilized for candidate store assignment, and MACS is adopted for support route planning. In addition, repair mechanisms and local improvement strategies are incorporated to maintain solution feasibility and enhance optimization performance.

The experimental results demonstrate that the proposed method effectively improves logistics route planning performance using real-world data. Compared with manually planned routes, the proposed framework reduces vehicle requirements, improves route loading efficiency, and enhances overall delivery performance. The ablation study comparing single-stage and multi-stage approaches verifies the effectiveness of the proposed framework, showing that multi-stage optimization improves solution quality and operational feasibility. Furthermore, evaluations on Solomon Benchmark Problems demonstrate competitive solution quality, particularly in reducing the number of required vehicles. These results indicate that the proposed framework effectively balances operational constraints and optimization objectives, providing practical value for real-world logistics applications.

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